$$
\begin{aligned}
&Y_u^{2\,i1\_i2\_} \to \frac{1}{16\,\pi^2} \Big(\tilde{\mu}\, y_e^{pr}\, y_u^{i1i2}\, \overline{a_e}^{pr}\, LF_{2,1,0}\big[m_{\tilde{l}}^{p},\, m_{\tilde{e}}^{r}\big] - \tilde{\mu}\, y_e^{pr}\, y_u^{i1i2}\, \overline{a_e}^{pr}\, LF_{3,1,-1}\big[m_{\tilde{l}}^{p},\, m_{\tilde{e}}^{r}\big] + \\
&\quad \frac{1}{2}\, y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e^{pr} + \overline{a_e}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_e^{pr} + 2\, \tilde{\mu}\, y_e^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{3,1,0}\big[m_{\tilde{l}}^{p},\, m_{\tilde{e}}^{r}\big] - \\
&\quad \frac{3}{2}\, y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e^{pr} + \overline{a_e}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_e^{pr} + 2\, \tilde{\mu}\, y_e^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{4,1,-1}\big[m_{\tilde{l}}^{p},\, m_{\tilde{e}}^{r}\big] + \\
&\quad y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_e}^{pr}\, y_e^{pr} + \overline{a_e}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_e^{pr} + 2\, \tilde{\mu}\, y_e^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{5,1,-2}\big[m_{\tilde{l}}^{p},\, m_{\tilde{e}}^{r}\big] + \\
&\quad 3\, \tilde{\mu}\, y_d^{pr}\, y_u^{i1i2}\, \overline{a_d}^{pr}\, LF_{2,1,0}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r}\big] - 3\, \tilde{\mu}\, y_d^{pr}\, y_u^{i1i2}\, \overline{a_d}^{pr}\, LF_{3,1,-1}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r}\big] + \\
&\quad \frac{3}{2}\, y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d^{pr} + \overline{a_d}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_d^{pr} + 2\, \tilde{\mu}\, y_d^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{3,1,0}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r}\big] - \\
&\quad \frac{9}{2}\, y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d^{pr} + \overline{a_d}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_d^{pr} + 2\, \tilde{\mu}\, y_d^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{4,1,-1}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r}\big] + \\
&\quad 3\, y_u^{i1i2} \Big(2\, \overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d^{pr} + \overline{a_d}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_d^{pr} + 2\, \tilde{\mu}\, y_d^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{5,1,-2}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r}\big] + \\
&\quad 3\, \tilde{\mu}\, y_u^{pr}\, y_u^{i1i2}\, \overline{a_u}^{pr}\, LF_{2,1,0}\big[m_{\tilde{u}}^{r},\, m_{\tilde{q}}^{p}\big] - 3\, \tilde{\mu}\, y_u^{pr}\, y_u^{i1i2}\, \overline{a_u}^{pr}\, LF_{3,1,-1}\big[m_{\tilde{u}}^{r},\, m_{\tilde{q}}^{p}\big] + \\
&\quad \frac{3}{2}\, y_u^{i1i2} \Big(\overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u^{pr} + 2\, \overline{a_u}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_u^{pr} + \tilde{\mu}\, y_u^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{3,1,0}\big[m_{\tilde{u}}^{r},\, m_{\tilde{q}}^{p}\big] - \\
&\quad \frac{9}{2}\, y_u^{i1i2} \Big(\overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u^{pr} + 2\, \overline{a_u}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_u^{pr} + \tilde{\mu}\, y_u^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{4,1,-1}\big[m_{\tilde{u}}^{r},\, m_{\tilde{q}}^{p}\big] + \\
&\quad 3\, y_u^{i1i2} \Big(\overline{C_{\phi1\phi2}}\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u^{pr} + 2\, \overline{a_u}^{pr} \Big(\overline{C_{\phi1\phi2}}\, a_u^{pr} + \tilde{\mu}\, y_u^{pr}\, (C_{\phi1^2} + C_{\phi2^2})\Big)\Big)\, LF_{5,1,-2}\big[m_{\tilde{u}}^{r},\, m_{\tilde{q}}^{p}\big] + \\
&\quad m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{2,1,0}[\tilde{\mu},\, m_1] - g_1^2\, y_u^{i1i2} \big(3\, \overline{C_{\phi1\phi2}} + m_1\, \tilde{\mu}\big)\, LF_{3,1,-1}[\tilde{\mu},\, m_1] + \\
&\quad m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{3,1,0}[\tilde{\mu},\, m_1] + 6\, \overline{C_{\phi1\phi2}}\, g_1^2\, y_u^{i1i2}\, LF_{4,1,-2}[\tilde{\mu},\, m_1] - \\
&\quad 3\, m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{4,1,-1}[\tilde{\mu},\, m_1] - \\
&\quad 3\, \overline{C_{\phi1\phi2}}\, g_1^2\, y_u^{i1i2}\, LF_{5,1,-3}[\tilde{\mu},\, m_1] + 2\, m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{5,1,-2}[\tilde{\mu},\, m_1] + \\
&\quad 3\, m_2\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, LF_{2,1,0}[\tilde{\mu},\, m_2] - 3\, g_2^2\, y_u^{i1i2} \big(3\, \overline{C_{\phi1\phi2}} + m_2\, \tilde{\mu}\big)\, LF_{3,1,-1}[\tilde{\mu},\, m_2] + \\
&\quad 3\, m_2\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{3,1,0}[\tilde{\mu},\, m_2] + 18\, \overline{C_{\phi1\phi2}}\, g_2^2\, y_u^{i1i2}\, LF_{4,1,-2}[\tilde{\mu},\, m_2] - \\
&\quad 9\, m_2\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{4,1,-1}[\tilde{\mu},\, m_2] - 9\, \overline{C_{\phi1\phi2}}\, g_2^2\, y_u^{i1i2}\, LF_{5,1,-3}[\tilde{\mu},\, m_2] + \\
&\quad 6\, m_2\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, (C_{\phi1^2} + C_{\phi2^2})\, LF_{5,1,-2}[\tilde{\mu},\, m_2] + \frac{2}{9}\, m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{1,1,1,0}\big[m_1,\, m_{\tilde{q}}^{i1},\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{1}{18}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,2,1,-1}\big[m_1,\, m_{\tilde{q}}^{i1},\, m_{\tilde{u}}^{i2}\big] + \\
&\quad \frac{1}{6}\, m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{1,1,1,0}\big[m_1,\, m_{\tilde{q}}^{i1},\, \tilde{\mu}\big] - \frac{1}{18}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big) \\
&\qquad LF_{2,2,1,-1}\big[m_1,\, m_{\tilde{u}}^{i2},\, m_{\tilde{q}}^{i1}\big] - \frac{2}{3}\, m_1\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{1,1,1,0}\big[m_1,\, m_{\tilde{u}}^{i2},\, \tilde{\mu}\big] - \\
&\quad \frac{1}{12}\, \overline{C_{\phi1\phi2}}\, g_1^2\, y_u^{i1i2}\, LF_{2,2,1,-2}\big[m_1,\, \tilde{\mu},\, m_{\tilde{q}}^{i1}\big] + \frac{1}{12}\, m_1\, C_{\phi2^2}\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{2,2,1,-1}\big[m_1,\, \tilde{\mu},\, m_{\tilde{q}}^{i1}\big] + \\
&\quad \frac{1}{3}\, \overline{C_{\phi1\phi2}}\, g_1^2\, y_u^{i1i2}\, LF_{2,2,1,-2}\big[m_1,\, \tilde{\mu},\, m_{\tilde{u}}^{i2}\big] - \frac{1}{3}\, m_1\, C_{\phi2^2}\, \tilde{\mu}\, g_1^2\, y_u^{i1i2}\, LF_{2,2,1,-1}\big[m_1,\, \tilde{\mu},\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{3}{2}\, m_2\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, LF_{1,1,1,0}\big[m_2,\, m_{\tilde{q}}^{i1},\, \tilde{\mu}\big] + \frac{3}{4}\, \overline{C_{\phi1\phi2}}\, g_2^2\, y_u^{i1i2}\, LF_{2,2,1,-2}\big[m_2,\, \tilde{\mu},\, m_{\tilde{q}}^{i1}\big] - \\
&\quad \frac{3}{4}\, m_2\, C_{\phi2^2}\, \tilde{\mu}\, g_2^2\, y_u^{i1i2}\, LF_{2,2,1,-1}\big[m_2,\, \tilde{\mu},\, m_{\tilde{q}}^{i1}\big] + \frac{8}{3}\, m_3\, \tilde{\mu}\, g_3^2\, y_u^{i1i2}\, LF_{1,1,1,0}\big[m_3,\, m_{\tilde{q}}^{i1},\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{2}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,2,1,-1}\big[m_3,\, m_{\tilde{q}}^{i1},\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{2}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,2,1,-1}\big[m_3,\, m_{\tilde{u}}^{i2},\, m_{\tilde{q}}^{i1}\big] + \\
&\quad \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2}\, \overline{a_d}^{pr}\, LF_{1,1,1,0}\big[m_{\tilde{d}}^{r},\, m_{\tilde{q}}^{p},\, \tilde{\mu}\big] + \frac{1}{2}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big) \\
&\qquad LF_{2,1,1,0}\big[m_{\tilde{d}}^{r},\, m_{\tilde{q}}^{p},\, \tilde{\mu}\big] - \frac{1}{2}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{d}}^{r},\, m_{\tilde{q}}^{p},\, \tilde{\mu}\big] - \\
&\quad \frac{1}{4}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big)\, LF_{2,2,1,-1}\big[m_{\tilde{d}}^{r},\, \tilde{\mu},\, m_{\tilde{q}}^{p}\big] + \\
&\quad \frac{1}{2}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big)\, LF_{2,1,1,0}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r},\, \tilde{\mu}\big] - \\
&\quad \frac{1}{2}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{q}}^{p},\, m_{\tilde{d}}^{r},\, \tilde{\mu}\big] - \\
&\quad \frac{1}{4}\, \tilde{\mu}\, y_d^{i1r}\, y_u^{pi2} \big(C_{\phi2^2}\, \overline{a_d}^{pr} + \overline{C_{\phi1\phi2}}\, \tilde{\mu}\, \overline{y_d}^{pr}\big)\, LF_{2,2,1,-1}\big[m_{\tilde{q}}^{p},\, \tilde{\mu},\, m_{\tilde{d}}^{r}\big] + \\
&\quad \frac{1}{9}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,1,1,0}\big[m_{\tilde{q}}^{i1},\, m_1,\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{1}{9}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{q}}^{i1},\, m_1,\, m_{\tilde{u}}^{i2}\big] + \\
&\quad \frac{4}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,1,1,0}\big[m_{\tilde{q}}^{i1},\, m_3,\, m_{\tilde{u}}^{i2}\big] - \\
&\quad \frac{4}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{q}}^{i1},\, m_3,\, m_{\tilde{u}}^{i2}\big] + \\
&\quad \frac{1}{9}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,1,1,0}\big[m_{\tilde{u}}^{i2},\, m_1,\, m_{\tilde{q}}^{i1}\big] - \\
&\quad \frac{1}{9}\, m_1\, g_1^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{u}}^{i2},\, m_1,\, m_{\tilde{q}}^{i1}\big] + \\
&\quad \frac{4}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{2,1,1,0}\big[m_{\tilde{u}}^{i2},\, m_3,\, m_{\tilde{q}}^{i1}\big] - \\
&\quad \frac{4}{3}\, m_3\, g_3^2 \big(\overline{C_{\phi1\phi2}}\, a_u^{i1i2} + C_{\phi2^2}\, \tilde{\mu}\, y_u^{i1i2}\big)\, LF_{3,1,1,-1}\big[m_{\tilde{u}}^{i2},\, m_3,\, m_{\tilde{q}}^{i1}\big]\Big)
\end{aligned}
$$